

MA 301

Test 2

Fall - 2011

Problem	points	out of
1		10
2		10
3		10
4		6
5		12
Score		48

Name: _____

HR: _____

Signature: _____

Formulas

* Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$

$$- z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

$$- \frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)] \quad (z \neq 0)$$

* Let $z = r(\cos \theta + i \sin \theta)$

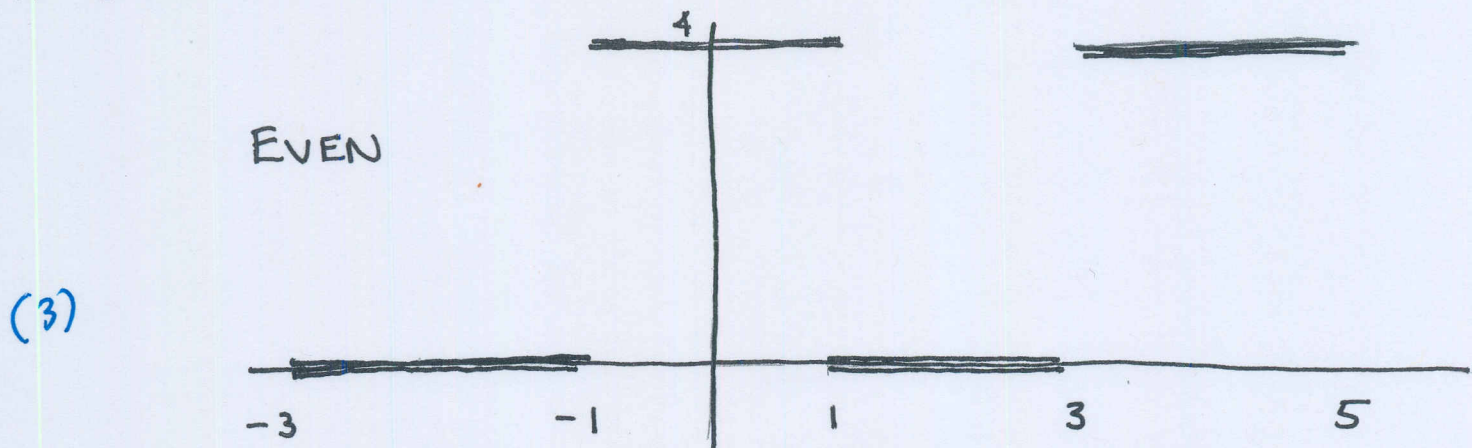
$$- z^n = r^n [\cos(n\theta) + i \sin(n\theta)]$$

$$- z^{1/n} = r^{1/n} \left[\cos \left(\frac{\theta + 2\pi k}{n} \right) + i \sin \left(\frac{\theta + 2\pi k}{n} \right) \right]$$

for $k = 0, 1, 2, \dots, n - 1$

Problem 1: Let $f(x) = \begin{cases} 0 & -2 < t < -1 \\ 4 & -1 < t < 1 \\ 0 & 1 < t < 2 \end{cases}$ and $f(x) = f(x+4)$

(a) Graph $f(x)$ on the interval $[-3, 5]$



(b) Find the Fourier Series of $f(x)$

(2) $b_n = 0$

(2) $a_0 = \frac{1}{4} \int_{-2}^2 f(x) dx = \frac{1}{4} (4)(2) = 2$

(2) $a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos\left(\frac{n\pi x}{2}\right) dx = \int_0^1 4 \cos\left(\frac{n\pi x}{2}\right) dx$

$$= 4 \left(\frac{2}{n\pi} \right) \sin\left(\frac{n\pi x}{2}\right) \Big|_0^1$$

$$= \frac{8}{n\pi} \left[\sin\left(\frac{n\pi}{2}\right) - \sin(0) \right]$$

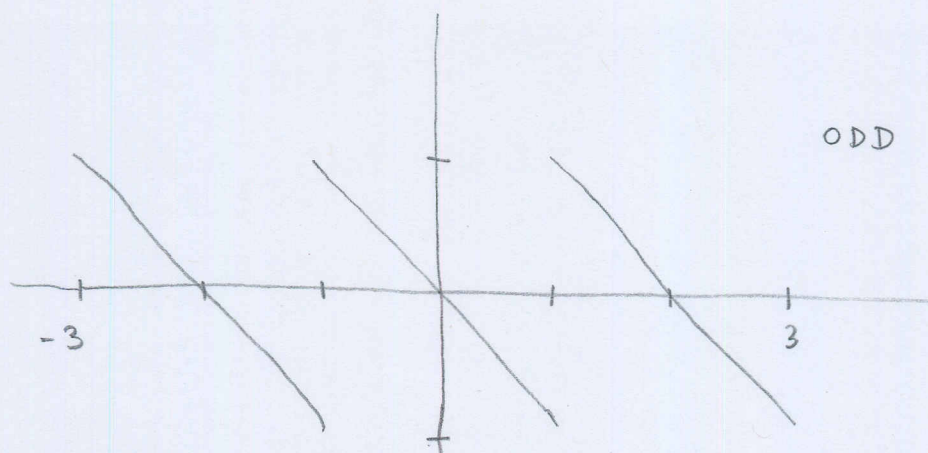
$$= \frac{8}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

(1) $f(x) = 2 + \sum_{n=1}^{\infty} \frac{8}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi x}{2}\right)$

Problem 2: Let $f(x) = -x$ on $[-1, 1]$ and satisfies $f(x) = f(x+2)$

$$L = 1$$

(a) Graph $f(x)$ on the interval $[-3, 3]$



(b) Find the Fourier Series of $f(x)$

$$\begin{aligned} u &= x & v &= -\frac{1}{n\pi} \cos(n\pi x) \\ du &= dx & dv &= \sin(n\pi x) \end{aligned}$$

$$a_0 = a_n = 0$$

$$b_n = \frac{1}{1} \int_{-1}^1 f(x) \sin(n\pi x) dx = 2 \int_0^1 -x \sin(n\pi x) dx$$

$$= -2 \left[-\frac{x}{n\pi} \cos(n\pi x) + \frac{1}{n\pi} \int_0^1 \cos(n\pi x) dx \right]$$

$$= 2 \left[\frac{x}{n\pi} \cos(n\pi x) - \frac{1}{n^2\pi^2} \sin(n\pi x) \right]_0^1$$

$$= \frac{2}{n\pi} \left[\left(\frac{1}{n\pi} \cos(n\pi) - \frac{1}{n^2\pi^2} \sin(n\pi) \right) - \left(0 - \frac{1}{n^2\pi^2} \sin(0) \right) \right]$$

$$= \frac{2}{n\pi} \left(\frac{1}{n\pi} \cos(n\pi) \right)$$

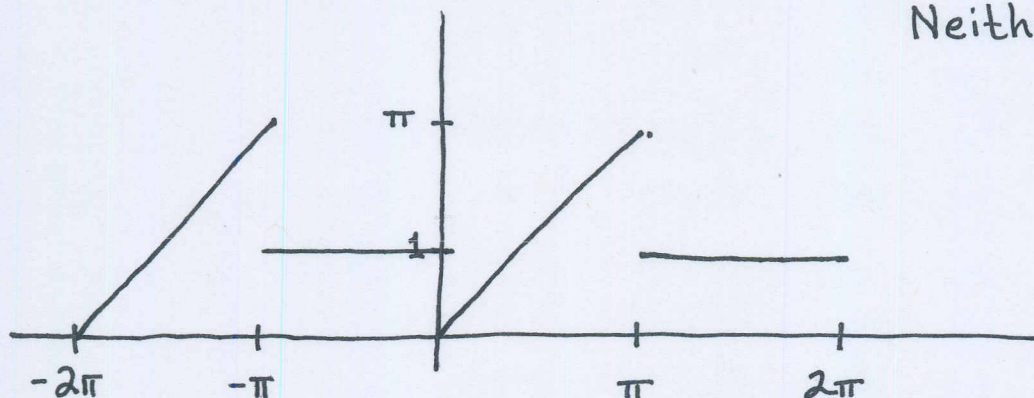
$$= \frac{2}{n^2\pi^2} \cos(n\pi)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} \cos(n\pi) \sin(n\pi x)$$

Problem 3: Let $f(x) = \begin{cases} 1 & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$ and $f(x) = f(x + 2\pi)$

$$L = \pi$$

(a) Graph $f(x)$ on the interval $[-2\pi, 2\pi]$



Neither Even nor
Odd

(b) Find the Fourier Series of $f(x)$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \left(\pi + \frac{1}{2}\pi^2 \right) = \frac{1}{2} + \frac{\pi}{4}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$\begin{aligned} u &= x & v &= \frac{1}{n} \sin(nx) \\ du &= dx & dv &= \cos(nx) dx \end{aligned}$$

$$= \frac{1}{\pi} \int_{-\pi}^0 \cos(nx) dx + \frac{1}{\pi} \int_0^{\pi} x \cos(nx) dx$$

$$= \frac{1}{\pi} \left(\frac{1}{n} \sin(nx) \right) \Big|_{-\pi}^0 + \frac{1}{\pi} \left[\frac{x}{n} \sin(nx) - \frac{1}{n} \int_0^{\pi} \sin(nx) dx \right]$$

$$= \frac{1}{\pi n} \left[x \sin(nx) + \frac{1}{n} \cos(nx) \right] \Big|_0^{\pi}$$

$$= \frac{1}{\pi n} \left[\left(\pi \sin(\pi n) + \frac{1}{n} \cos(\pi n) \right) - \left(0 + \frac{1}{n} \cos(0) \right) \right]$$

$$= \frac{1}{\pi n^2} \cos(\pi n) - \frac{1}{\pi n^2}$$

$$= \frac{1}{\pi n^2} (1 - \cos(\pi n))$$

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$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 \sin(nx) dx + \frac{1}{\pi} \int_0^{\pi} x \sin(nx) dx$$

$$= \frac{1}{\pi} \left(-\frac{1}{n} \cos(nx) \right) \Big|_{-\pi}^0 + \frac{1}{\pi} \left(-\frac{x}{n} \cos(nx) + \frac{1}{n} \int_0^{\pi} \cos(nx) dx \right)$$

$$= \frac{1}{\pi n} \left(\cos(0) - \cos(-\pi n) \right) + \frac{1}{\pi n} \left[-x \cos(nx) + \frac{1}{n} \sin(nx) \right]_0^{\pi}$$

$$= -\frac{1}{\pi n} (1 - \cos(\pi n)) + \frac{1}{\pi n} \left[(-\pi \cos(\pi n) + \frac{1}{n} \sin(\pi n)) - \left(0 + \frac{1}{n} \sin(0) \right) \right]$$

$$= -\frac{1}{\pi n} (1 - \cos(\pi n)) + \frac{1}{\pi n} (-\pi \cos(\pi n))$$

$$= -\frac{1}{\pi n} (1 - \cos(\pi n)) - \frac{1}{n} \cos(\pi n)$$

$$= -\frac{1}{\pi n}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

$$u = x$$

$$du = dx$$

$$v = \frac{1}{n} \cos(nx)$$

$$dv = \sin(nx) dx$$

Perform the following operations and write the result in standard form.

$$(a) \quad z_1 + z_2 = 5 + i + 2 - 4i = 7 - 3i$$

$$(b) \quad z_1 z_3 = (5 + i)(a + bi) = 5a + 5bi + ai + bi^2 \\ = 5a - b + (5b + a)i$$

$$(c) \quad \frac{z_2}{z_1} = \frac{(2 - 4i)(5 - i)}{(5 + i)(5 - i)} = \frac{10 - 4 + (-2 - 20)i}{25 + 1} = \frac{6}{26} - \frac{22}{26}i$$

$$(d) \quad |z_2| = |2 - 4i| = \sqrt{4 + 16} = \sqrt{20}$$

$$(e) \quad z_1 - iz_2 = 5 + i - i(2 - 4i) = 5 + i - 2i - 4 \\ = 1 - i$$

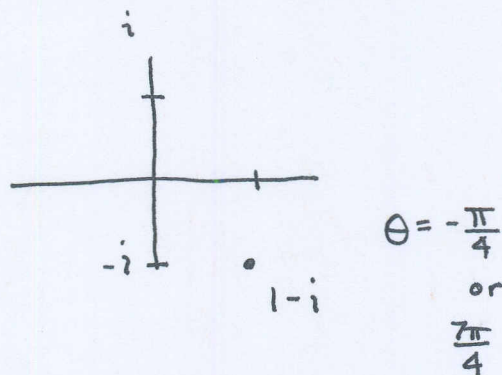
$$(f) \quad \sqrt{-10} \cdot \sqrt{-6} = 10i \cdot 6i = 60i^2 = -60$$

Problem 5: Let $z_1 = 1 - i$, $z_2 = 3i$. Answer the following:

(a) Find the polar form of z_1 and plot it.

$$r = |z_1| = \sqrt{2} \quad \theta = -\frac{\pi}{4}$$

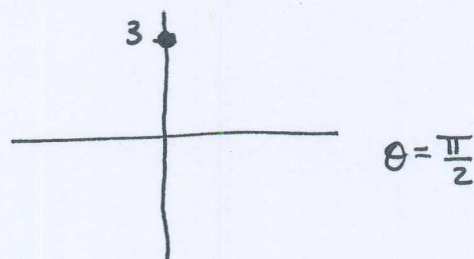
$$z_1 = \sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$$



(b) Find the polar form of z_2 and plot it.

$$r = |z_2| = 3$$

$$z_2 = 3 \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right)$$



(c) Find z_1^{12} .

$$\begin{aligned} z_1^{12} &= (\sqrt{2})^{12} \left(\cos(-3\pi) + i \sin(-3\pi) \right) \\ &= 2^6 \left(\cos(3\pi) - i \sin(3\pi) \right) = -64 \end{aligned}$$

(d) Find $z_1^{1/3}$.

$$n=3 \quad z_1^{1/3} = (\sqrt{2})^{1/3} \left(\cos\left(\frac{-\frac{\pi}{4} + 2\pi k}{3}\right) + i \sin\left(\frac{-\frac{\pi}{4} + 2\pi k}{3}\right) \right)$$

for $k=0, 1, 2$

3 values

(d) Solve for z : $z^4 + z_1 = z_2$

$$z^4 + 1 - i = 3i \Rightarrow$$

$$z^4 = -1 + 4i$$

$$\Rightarrow z = (-1 + 4i)^{1/4}$$

$$\begin{aligned} \theta &= \tan^{-1}(4) \\ r &= \sqrt{17} \end{aligned}$$

$$= (\sqrt{17})^{1/4} \left(\cos\left(\frac{\theta + 2\pi k}{4}\right) + i \sin\left(\frac{\theta + 2\pi k}{4}\right) \right)$$

for $k=0, 1, 2, 3$